

Topic No 10

Schrodinger Equation :-

Schrodinger derived an equation based on de-Broglie dual behaviour of electrons and Heisenberg uncertainty Principle is known as Schrodinger's Wave equation.

Introduction

The equation is named after Erwin Schrodinger who postulated the equation in 1925 and published in 1926. The work that resulted in his Nobel Prize in Physics in 1933.

Applications

⇒ To explain the behaviour of micro particle.

⇒ To determine the total energy.

⇒ To determine the translational energy.

⇒ To analyse molecular motion.

Derivation

Classical Physics

$$\nabla^2 \phi = \frac{1}{v^2} \frac{\partial^2 \phi}{\partial t^2} \quad \text{--- (1)} \quad \nabla^2 \phi = ?$$

We know that $\nabla^2 = \left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right]$
xly ϕ on b.s.

$$\nabla^2 \phi = \left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right] \phi \Rightarrow \nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} \quad \text{--- (2)}$$

Then $\phi = (x, y, z, t) = \psi(x, y, z) e^{iwt}$

only x-axis.

$$\nabla^2 \psi = \frac{\partial^2 \psi}{\partial x^2} \quad \text{apply derivative w.r.t. x.} \quad \lambda = \frac{h}{\sqrt{2m(E-V_0)}} \quad (7)$$

again derivative

$$\frac{\partial \psi}{\partial x} = \frac{\partial \psi}{\partial x} e^{i\omega t}$$

$$\frac{\partial^2 \psi}{\partial x^2} = \frac{\partial^2 \psi}{\partial x^2} e^{i\omega t} \quad (3)$$

Then wave = $v = f\lambda$

$$\Rightarrow \frac{v}{f} = \frac{h}{\sqrt{2m(E-V_0)}} \quad (8)$$

Derivative w.r.t. time.

eq No 8 apply reciprocal.

$$\psi(x, y, z, t) = \psi(x, y, z) e^{i\omega t}$$

$$\frac{\partial \psi}{\partial t} = \psi e^{i\omega t} i\omega$$

again diff.

$$\frac{\partial^2 \psi}{\partial t^2} = \psi e^{i\omega t} (i\omega)(i\omega)$$

$$\frac{\partial^2 \psi}{\partial t^2} = -\psi \omega^2 e^{i\omega t} \quad (4)$$

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To find $\frac{1}{v^2} = ?$

When body move so gained KE & P.E.

Total energy $x \lambda$

$$E = K.E + V_0$$

$$E = \frac{1}{2}mv^2 + V_0$$

$$E - V_0 = \frac{1}{2}mv^2 \Rightarrow 2(E - V_0) = mv^2$$

xlym on b.s

$$2m(E - V_0) = m^2 v^2 \quad (5)$$

Know that de-broglie eq

$$\lambda = \frac{h}{mv} \Rightarrow \lambda^2 = \frac{h^2}{m^2 v^2}$$

$$\lambda^2 = \frac{h^2}{2m(E - V_0)} \quad \text{Taking square root on b.s}$$

$$F = \sqrt{2m(E - V_0)}$$

$$\frac{1}{v} = \frac{h}{\sqrt{2m(E - V_0)}} \quad hf$$

We know that $\frac{1}{v} = \frac{1}{v^2}$

$$\frac{1}{v^2} = \frac{(2m(E - V_0))}{h^2 f^2} \quad (9)$$

Putting value in eq (1)

$$\nabla^2 \psi = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}$$

Now putting eq (3, 4, 9)

$$\Rightarrow \frac{\partial^2 \psi}{\partial x^2} e^{i\omega t} = \left(\frac{2m(E - V_0)}{h^2 f^2} \right) (-\omega^2 \psi e^{i\omega t})$$

$$\frac{\partial^2 \psi}{\partial x^2} = \left(\frac{2m(E - V_0)}{h^2 f^2} \right) x - \omega^2 \psi$$

$$\omega = 2\pi f \quad \text{putting value in } \omega^2$$

$$= \frac{4\pi^2 f^2 (2m(E - V_0))}{h^2 f^2} \psi$$

$$\frac{\partial^2 \psi}{\partial x^2} = \frac{8\pi^2 m}{h^2} (E - V_0) \psi$$

$$k = \frac{h}{\lambda} \Rightarrow h = 2\pi \hbar$$

$$h^2 = 4\pi^2 \hbar^2$$

$$\frac{d^2 \psi}{dx^2} = -\frac{2m(E - V_0)}{\hbar^2} \psi$$

Putting value of $\hbar^2$

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} = (E - V_0) \psi \Rightarrow \frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} = E\psi - V_0\psi$$

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} + V_0\psi = E\psi \Rightarrow \left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V_0 \right] \psi = E\psi$$

Hamiltonian $\leftarrow$

$$H = T + P.E \Rightarrow E = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V_0$$

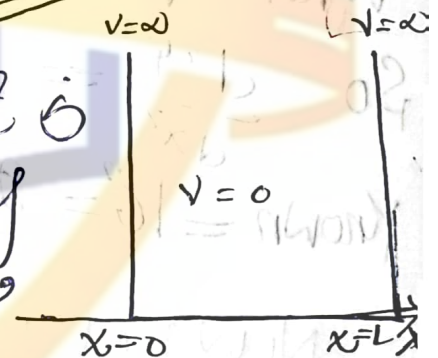
$$H\psi = E\psi$$

This is called Schrodinger Wave equation.

* Topic No 12

Schrodinger Wave equation in one Dimensional.

* Consider a Particle that is constrained to move only in x direction from $x=0$ to $x=L$.



Case No 1

* The Particle inside a box the Potential energy is equal to zero.

Apply Schrodinger Wave equation

$$\frac{d^2 \psi}{dx^2} = -\frac{2m(E - V)}{\hbar^2} \psi$$

Case No 2

While other side the Particle Present outside and inner side of wall the Potential energy is ∞

Derivation: $\hat{H}\psi = E\psi$ — (1)

$\hat{H}$ = Hamiltonian operator

in x direction: $\Rightarrow \hat{H} = \frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V$

Putting value in eq no 1

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V\psi = E\psi \Rightarrow \frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} - V\psi + E\psi = 0$$

NOW $-\frac{2m}{\hbar^2}$ xly on b.s

$$\left(\frac{\hbar^2}{2m}\right) \left(\frac{2m}{\hbar^2}\right) \frac{d^2\psi}{dx^2} = (E - V)\psi \left(\frac{2m}{\hbar^2}\right) \Rightarrow \frac{d^2\psi}{dx^2} = -\frac{2m}{\hbar^2} (E - V)\psi$$

When $V = 0$

$$\text{So } \frac{d^2\psi}{dx^2} = -\frac{2mE}{\hbar^2} \psi \Rightarrow \frac{d^2\psi}{dx^2} + \frac{2mE}{\hbar^2} \psi = 0$$

$$\text{Known } = k^2 = \frac{2mE}{\hbar^2} \quad (3)$$

NOW Pulling Value k in eq no 2

$$\frac{d^2\psi}{dx^2} + k^2\psi = 0 \quad \text{Apply 2nd order differential equation}$$

$$\text{Now a } \psi(x) = A \sin kx + B \cos kx \quad (4)$$

The A, B are the arbitrary constant.

$$\text{So } x=0 \quad \psi(0)=0 \quad \text{Putting eq (4)}$$

$$0 = A \sin 0 + B \cos 0$$

$$0 = 0 + B \Rightarrow \boxed{B=0} \quad \because \begin{matrix} \sin 0 = 0 \\ \cos 0 = 1 \end{matrix}$$

When $x = L$

$$\psi(L) = 0$$

Particle not present in the wall it is not true.

By putting this value in eq NO (4)

$$0 = A \sin KL + 0 \Rightarrow A \sin KL = 0$$

$$\sin KL = 0 \quad \therefore KL = n\pi \quad (\pi, 2\pi, 3\pi \dots)$$

$$KL = n\pi \Rightarrow K = \frac{n\pi}{L} \quad \text{but } K^2 = \frac{2mE}{\hbar^2}$$

$$\text{So } K^2 = \frac{n^2\pi^2}{L^2} \quad \text{Taking sq on b.s } K^2 = \frac{n^2\pi^2}{L^2} \quad (5)$$

Comparing eq NO (3) & (5)

$$\therefore \hbar = \frac{h}{2\pi}$$

$$\frac{2mE}{\hbar^2} = \frac{n^2\pi^2}{L^2} \quad \text{Now putting value } \hbar^2 = \frac{h^2}{4\pi^2}$$

$$\frac{2mE}{h^2} \times 4\pi^2 = \frac{n^2\pi^2}{L^2} \Rightarrow \boxed{\frac{h^2 n^2}{8mL^2} = E}$$

So this formula the energy give the Particle in one dimension box, Energy of Particle is Quantized.

The Probability finding the Particle in a small space b/w x & $x+dx$ is given by $\psi^2(x)dx$.

NOW $\boxed{\psi = A \sin Kx} \Rightarrow A \sin\left(\frac{n\pi}{L}x\right)$

$$\int_0^L \psi^2(x) dx = 1 \Rightarrow \int_0^L A^2 \sin^2(Kx) dx = 1$$

$$A^2 \int_0^L \frac{1 - \cos 2Kx}{2} dx = 1 \Rightarrow \frac{A^2}{2} \int_0^L [1 - \cos 2Kx] dx = 1$$

$$\frac{A^2}{2} \left[x \Big|_0^L - \frac{1}{2K} \sin 2Kx \Big|_0^L \right] = 0$$

$$\frac{A^2}{2} \left[L - \frac{1}{2K} \sin 2KL \right] \Rightarrow \frac{A^2}{2} \left[L - \frac{1}{2K} \sin \frac{n\pi x}{L} \right]$$

$$\frac{A^2}{2} \left[L - \frac{1}{2K} \sin n\pi \right] = 1 \quad \sin n\pi = 0$$

$$\frac{A^2}{2} [L] = 1, \quad A^2 = \frac{2}{L} \Rightarrow A = \sqrt{\frac{2}{L}}$$

Now Putting value A in eq no (6)

$$\psi = \sqrt{\frac{2}{L}} \sin \left[\frac{n\pi}{L} \right] x$$

$\Rightarrow$ Zero point energy:

Electron is not at rest even at zero K temp. The minimum energy is called Zero point energy.

When $n=1$

$$E_1 = \frac{h^2}{8ma^2}$$

Linearity and Superposition Principle: $\frac{d\psi}{dx}, \frac{d\psi}{dt}, \frac{d\psi}{dz}$

* The important Property of Schrodinger's equation is that it is linear in wave function ψ . This means that the eqn is in terms of ψ and its derivative and no term is independent of ψ .

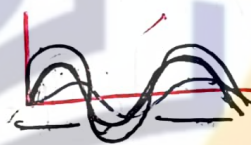
$\Rightarrow$ Thus the Linear Property is defined as if $\psi_1, \psi_2, \psi_3, \psi_4$ — are different wave functions representing different quantum states of the same system then their linear combination with complex coefficients will represent the wave function of a given system. □

$$\psi = C_1\psi_1 + C_2\psi_2 + C_3\psi_3 + \dots$$

It is also a solution of Schrodinger equation.

$$\psi = \sum_{i=1}^n C_i \psi_i$$

Superposition Principle:



$$a_1\psi_1 + a_2\psi_2$$

When two or more waves moving through the same region of space, waves will superimpose and produce a well-defined combined effect. $\rightarrow$ If $\psi = a_1\psi_1 + a_2\psi_2$ is a solution.

here $a_1, a_2 = \text{constant}$. Then, ψ_1, ψ_2 follow the Superposition Principle. That means the net amplitude caused by two or more waves travelling the same space is the sum of amplitudes.

that would have been produced by the individual waves separately.

Mathematical equations:

If ψ_1 and ψ_2 are wave function and $\hat{O}$ is operator.

$$\text{Then } \hat{O}\psi_1 = \lambda \psi_1 \quad \text{--- (1)}$$

$$\hat{O}\psi_2 = \lambda \psi_2 \quad \text{--- (2)}$$

Linear combination.

$$\psi = c_1 \psi_1 + c_2 \psi_2 \quad \text{--- (3)}$$

Apply $\hat{O}$ on both sides

$$\hat{O}\psi = \hat{O}(c_1 \psi_1 + c_2 \psi_2)$$

$$\hat{O}\psi = \hat{O}c_1 \psi_1 + \hat{O}c_2 \psi_2 \Rightarrow \hat{O}\psi = \lambda c_1 \psi_1 + \lambda c_2 \psi_2$$

$$\hat{O}\psi = \lambda (c_1 \psi_1 + c_2 \psi_2)$$

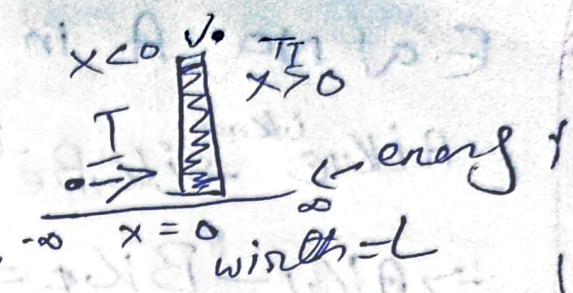
$$\boxed{\hat{O}\psi = \lambda \psi} \rightarrow \text{Superposition Principle}$$

Topic No. 9 Barrier tunneling / Step Potential

اس کے ایک حصے میں ہم نے دیکھا ہے کہ کسی بھی چھوٹے بارئیر (Barrier) میں کتنی کم احتمال (Probability) ہے کہ وہ واپس آئے گا یا نہ اسے ٹھیک سے آگے چلا جائے گا اسے بارئیر میں نہ آئے

کے ہیں۔ Step Potential / Barrier tunneling

انٹر پاد ٹیبل کی انری ہلکم یوگی تو وہ Barrier سے ٹکرا کر واپس آئے گا اور انٹر انری زیادہ یوگی تو وہ آگے چلا جائے گا



Derivation

For Region I

$V(x) = 0$ is called energy

use Schrodinger wave eq

$$\left[\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V(x) \right] \psi = E \psi$$

$$\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V(0)\psi = E \psi$$

$$\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E \psi \Rightarrow \frac{d^2\psi}{dx^2} = \frac{E \psi}{\frac{\hbar^2}{2m}}$$

$$\frac{d^2\psi}{dx^2} = -\frac{2m}{\hbar^2} E \psi \Rightarrow \frac{d^2\psi}{dx^2} + \frac{2mE}{\hbar^2} \psi = 0$$

$K_1 = \frac{2m}{\hbar^2}$ is constant

$$\frac{d^2\psi}{dx^2} + K_1^2 \psi = 0 \quad (1)$$

Using 2nd order d.e.v

$$\frac{d^2\psi}{dx^2} + K_1^2 \psi = 0$$

$$\psi_1 = A e^{iK_1 x} + B e^{-iK_1 x} \quad (2)$$

For Region II

$$\frac{\hbar^2}{2m} \frac{d^2\psi_2}{dx^2} + V(x)\psi_2 = E \psi_2$$

$V(x) = V_0$ Then

$$\frac{d^2\psi_2}{dx^2} + \frac{2m}{\hbar^2} V_0 \psi_2 = \frac{2m}{\hbar^2} E \psi_2$$

$$\frac{d^2\psi_2}{dx^2} + (E - V_0) \frac{2m}{\hbar^2} \psi_2 = 0$$

$$\frac{d^2\psi_2}{dx^2} + K_2^2 \psi_2 = 0 \quad (3)$$

$$\psi_2(x) = C e^{iK_2 x} \quad (4)$$

- A $\rightarrow$ incident particle behavior
- B $\rightarrow$ Reflected particle behavior
- C $\rightarrow$ Transmitted particle behavior

Boundary condition

at $x = 0$

$$\psi_1(x) = \psi_2(x)$$

$$A e^{iK_1 x} + B e^{-iK_1 x} = C e^{iK_2 x} \quad (5)$$

$$A + B = C \quad (6)$$

$$\frac{d\psi_1(x)}{dx} = \frac{d\psi_2(x)}{dx}$$

Eqn No. A in differential eq:

$$A i k_1 e^{i k_1 x} - i k_1 B e^{-i k_1 x} = i k_2 C e^{i k_2 x} \Rightarrow A i k_1 e^{i k_1(0)} - B i k_1 e^{-i k_1(0)} = i k_2 C e^{i k_2(0)}$$

$$\Rightarrow A k_1 - B k_1 = C k_2 \quad \text{--- (6)}$$

Putting value c
in eq No 6

$$A k_1 - B k_1 = (A + B) k_2$$

$$k_1 (A - B) = (A + B) k_2$$

$$A k_1 - B k_1 = A k_2 + B k_2 \quad \therefore \text{arrange same value}$$

$$A k_1 - A k_2 = B k_2 + B k_1 \Rightarrow \text{common A \& B}$$

$$A (k_1 - k_2) = B (k_2 + k_1) \Rightarrow A \frac{(k_1 - k_2)}{k_1 + k_2} = B \quad \text{--- (7)}$$

We know that $C = A + B$

$$C = A + \left(\frac{k_1 - k_2}{k_1 + k_2} \right) A \quad \text{Now taking LCM}$$

$$C = \frac{k_1 + k_2 + k_1 - k_2}{k_1 + k_2} A \Rightarrow \boxed{C = \frac{2 k_1 A}{k_1 + k_2}} \quad \text{--- (8)}$$

$$\psi_1(x) = A e^{i k_1 x} + B e^{-i k_1 x}$$

Putting value B
in eq No 2

$$\psi_1(x) = A e^{i k_1 x} + \left[\frac{k_1 - k_2}{k_1 + k_2} \right] A e^{-i k_1 x} \quad \text{--- (9) with } \psi$$

$$\psi_2(x) = \frac{2 k_1}{k_1 + k_2} A e^{i k_2 x} \quad \text{--- (10)}$$

* Find the Probability of Reflection and Transmitted Particles

$$\Rightarrow \text{flux} = \int \psi^* \hat{v} \psi dx$$

For incident Particl- $\Rightarrow \psi_i(x) = Ae^{ik_1x}$

For Reflection // $\Rightarrow \psi_r(x) = Be^{-ik_2x}$

For Transmitted // $\Rightarrow \psi_T(x) = Ce^{ik_2x}$

*** Reflection**

$R = \frac{\text{Flux of Probability}}{\text{Incident Flux}}$

$$\therefore \hat{p} = mv \Rightarrow \frac{j\hbar}{m} \frac{\partial}{\partial x} = \hat{v}$$

$$R = \frac{\int_{-\infty}^{\infty} \psi^* \hat{v} \psi dx}{\int_{-\infty}^{\infty} \psi^* \hat{v} \psi dx}$$

As we know that ψ^* mean conjugate.

$$\psi_i^* = Be^{ik_1x}, \psi_r = Be^{-ik_2x}$$

Putting value in formula

$$R = \frac{\int_{-\infty}^{\infty} Be^{ik_1x} \frac{j\hbar}{m} \frac{\partial}{\partial x} Be^{-ik_2x} dx}{\int_{-\infty}^{\infty} Ae^{-ik_2x} \frac{j\hbar}{m} \frac{\partial}{\partial x} Ae^{ik_1x} dx}$$

$$R = \frac{B^2 \int e^{ik_1x} -ik_2x e^{-ik_2x} dx}{A^2 \int e^{ik_1x} ik_1x e^{ik_1x} dx}$$

$$R = \frac{B^2}{A^2} \quad \text{Putting value } B \text{ \& } A$$

$$\frac{(k_1 - k_2)^2}{(k_1 + k_2)^2} \div A^2 \times A^2$$

$$R = \frac{(k_1 - k_2)^2}{(k_1 + k_2)^2}$$

Transmission

Same formula

$$T = \frac{\int Ce^{-ik_2x} \frac{j\hbar}{m} \frac{\partial}{\partial x} Ce^{ik_2x} dx}{\int Ae^{ik_1x} \frac{j\hbar}{m} \frac{\partial}{\partial x} Ae^{ik_1x} dx}$$

$$T = \frac{k_2^2 C^2}{k_1 A^2} \quad \text{Putting value } C \text{ and } A$$

$$T = \frac{(2k_1)^2}{(k_1 + k_2)^2} \div k_1 A^2$$

$$T = \frac{2k_1 k_2 A^2}{(k_1 + k_2)^2} \div A^2$$

$$T = \frac{4k_1 k_2}{k_1 + k_2}$$

k independent on mass.

$$T \propto \frac{1}{R} = 1$$

$$R + T = 1$$